

IMPLEMENTED

# Upgrade Tracker

Performance + visual system refresh

A faster, calmer tracker with shared DevKit primitives, import-first onboarding, visible data freshness, and plan-derived village completion dates.

6

performance changes

7

tracker design  
primitives

5

village cost/time  
sections

# Six changes aimed at scroll smoothness

Work is now paid only when the visible UI needs it.

1

## Virtualized grids

Wrap + shrinkWrap eagerly built every upgrade and collection tile. SliverGrid.builder now creates visible cells on demand.

2

## Local search state

Search, sort, and filters moved into their tabs, so typing no longer rebuilds the entire tracker shell and every page.

3

## Cached derived data

Item filters, summaries, static lookups, and collection indexes are memoized instead of rescanned during routine rebuilds.

4

## Scoped ticking

A ValueNotifier updates only active timers and the freshness label; the root page no longer setState every second.

5

## Cheaper image/audio work

Owned art avoids ColorFiltered, scenery audio uses disk loading, and its position timer stops whenever playback stops.

6

## Deferred secondary work

Imported snapshots are parsed once, reused from cache, and widget synchronization waits for the first frame at idle priority.

# A semantic style sheet for tracker UI

Roles are named by purpose, so screens stop inventing weights, colors, and density.

TYPE ROLES

Hero metric

32 / 800

Dynamic Type is inherited from Flutter TextTheme.  
Compact controls keep a 44pt minimum and grow when text scales.

Screen title

24 / 700

Section title

17 / 700

Compact · 44pt

Standard · 52pt

Row title

15 / 600

Body

15 / 500

Metadata

12 / 500

STATE COLORS



Builders



Research



Pets



Complete



Scheduled



Unavailable

MATERIAL RULE

Glass for focus

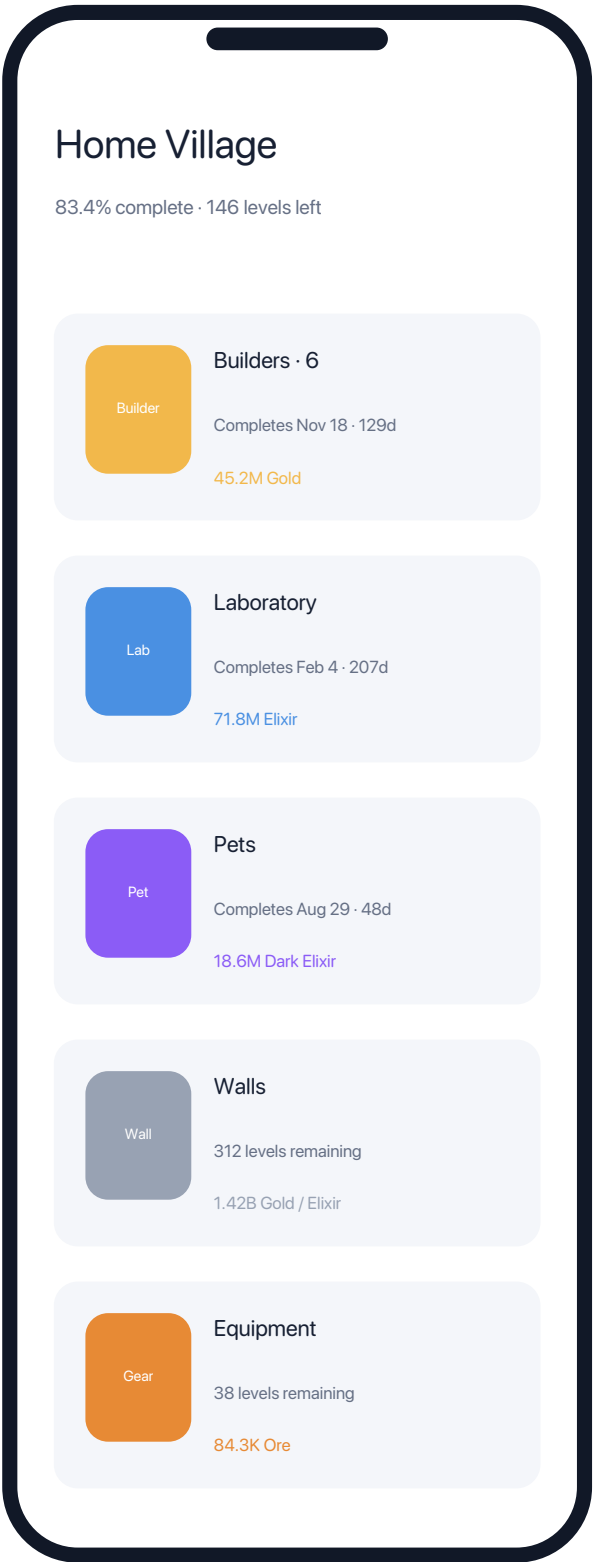
Hero and floating controls only.

Quiet sections

Lists use low-contrast CKSectionPanel.

# Completion dates and costs, by queue

Plan math powers finish dates; walls and equipment stay explicitly untimed.

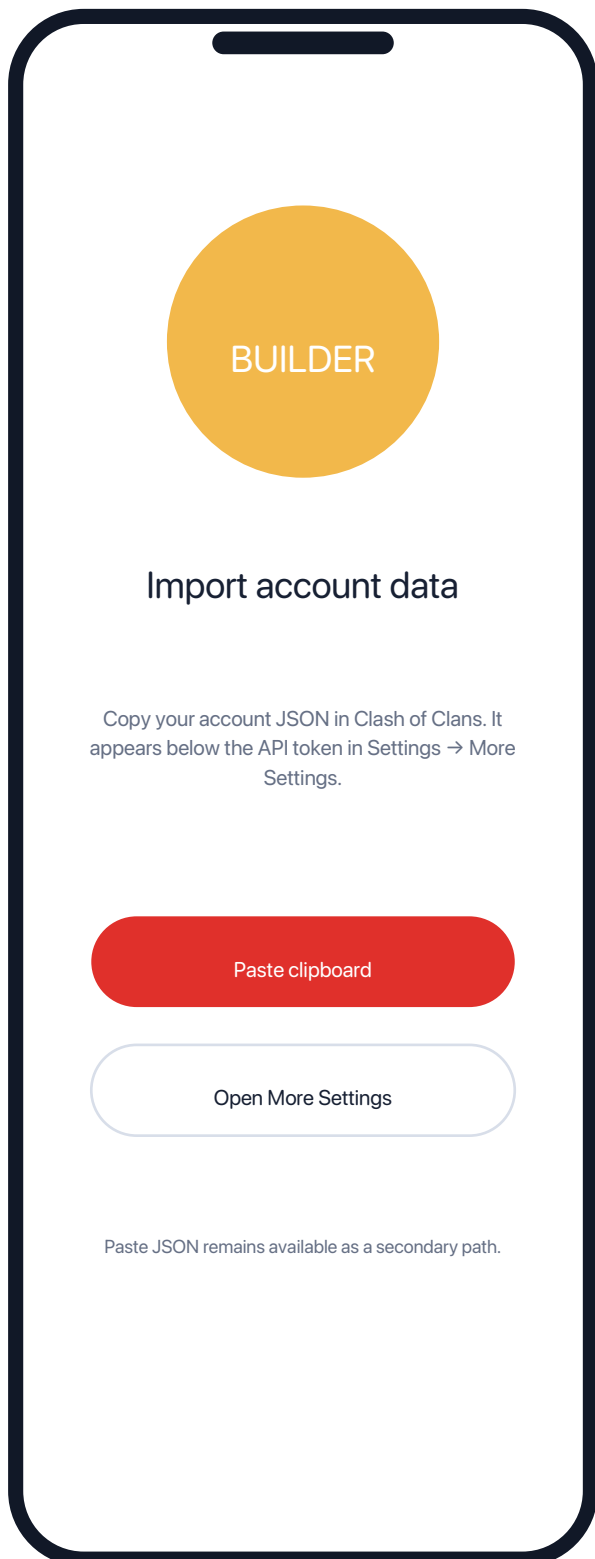


ACCOUNTING RULES

- Builder finish**  
Uses the same plan lanes as Plan, including the actual builder count for Home Village or Builder Base.
- Lab + pets**  
Each queue has its own completion date, duration, and resource totals; pets appear only for Home Village.
- Walls + equipment**  
Shown as explicit cost sections with no fabricated date because their upgrades have no time.
- Builders + helpers**  
Visible under their villages, but excluded from Home Village and Builder Base overall completion.
- Game artwork**  
Every row selects the relevant Clash asset; no generic category glyphs substitute for game content.

# A useful first screen before JSON exists

The tracker explains the source, opens Clash directly, and makes snapshot age visible.



## FRESHNESS SURFACES

### Magic Jr.

#2J8V28GV0 · TH18 / BH10

Updated 18 minutes ago

### Account picker

Every saved account shows its capture date/time, so stale exports are visible before selection.

### Live age label

Only the small timestamp listens to the clock. The page and its grids stay still.

VERIFICATION

# What changed, and how it was checked

App and shared system were validated independently.

|              |                                 |           |
|--------------|---------------------------------|-----------|
| ClashKingApp | Tracker parser/repository tests | 6 passed  |
| ClashKingApp | Tracker source analysis         | No issues |
| ClashKingApp | Localization generation         | Generated |
| DevKit       | Flutter component tests         | 9 passed  |
| DevKit       | Manifest + token checks         | Passed    |
| Both repos   | Whitespace / patch validation   | Clean     |

IMPLEMENTATION MAP

|                                 |                                                               |
|---------------------------------|---------------------------------------------------------------|
| upgrade_tracker_page.dart       | virtualized tabs, scoped timers, onboarding, freshness, modal |
| upgrade_tracker_models.dart     | memoized item queries and summaries                           |
| upgrade_tracker_parser.dart     | cached static lookup and O(1) collection dedupe               |
| upgrade_tracker_repository.dart | snapshot cache, one-pass imports, capture metadata            |
| clashking_design_system.dart    | semantic type, color, density, and tracker primitives         |